

How Have Coffee Consumption Impacted NYC High Students' Academic Performance and
Productivity?

AP Research 2025

Word Count: 3746

Introduction

According to the National Health and Nutrition Examination Survey (NHANES), around 154 million adults, or 75% of the US population, reported drinking coffee, with 49% consuming it daily (Loftfield et al., 2016). Coffee, one of the most widely consumed beverages in the world, is often praised for its ability to enhance alertness and cognitive function. The primary psychoactive compound in coffee, caffeine can enhance mental alertness, memory, and cognitive performance (Cano-Marquina et al., 2013). These effects may benefit students, particularly high school students, who face increasing academic demands and pressures to perform well. In particular, students may turn to coffee as a means to boost their energy levels, cognitive abilities, and overall productivity during long study sessions.

For the purposes of this study, coffee consumption refers to the intake of coffee by high school students, while academic performance is assessed by Grade Point Average (GPA) and cognitive efficiency in completing schoolwork. Productivity is measured by the quantity and quality of schoolwork completed, while sleep patterns refer to the duration and quality of sleep on school nights. Moderate coffee consumption is defined as 1-4 cups of coffee per day, providing between 80-400 milligrams of caffeine.

Previous research has found that moderate caffeine consumption can support improvements in attention, alertness, and memory. Santos et al. (2010) suggest that caffeine enhances focus and mental alertness, leading to improved academic performance. However, other studies have shown that excessive intake can lead to risks of anxiety, jitteriness, and sleep disturbances (Pereira et al, 2019). Ahsan F and Bashir S (2019) further emphasize that consumption in large amounts is linked with development of heart diseases, stroke, cardiac

arrhythmias, and hypertension. However, the effects of coffee consumption vary depending on individual tolerance, lifestyle choices, and time of consumption.

Despite the growing popularity of caffeine among high school students, few studies have focused specifically on how coffee consumption affects their academic performance and productivity levels. High school students may metabolize caffeine differently from adults and are particularly vulnerable to sleep disruption, making the group an important population to study. This elicits the question: To what extent does coffee consumption affect the academic performance and productivity levels of high school students in NYC? This study seeks to explore the academic success of high school students and provide recommendations to help address the impacts of coffee consumption on NYC high school students.

Literature Review

Cognitive Effects and Academic Success

A systematic review and meta-analysis conducted by Santos et al. (2010) found that there is a correlation between higher caffeine intake and a reduced risk of cognitive impairment. While their finding mainly targets older populations, it suggests that the implication that caffeine supports neurological function could be relevant for high school students engaging in demanding coursework. This finding suggested that coffee consumption could play a positive role against cognitive decline.

Supporting caffeine's cognitive benefits, Haskell-Ramsay et al. (2018) found participants who consumed regular coffee exhibited faster reaction times and improved attention accuracy compared to participants who consumed decaffeinated coffee or placebo. Haskell-Ramsay et al.

also found that decaffeinated coffee also led to an increase in alertness compared to placebo, suggesting that non-caffeine compounds like chlorogenic acids may also play a role in cognitive performance. This implies that decaf could be a potential alternative with some cognitive upsides found in coffee.

Similarly, Raquel and Jesus (2016) conducted with the group exposed to volatile coffee outperforming the control group in recall tasks. They found that students who consumed coffee performed better in short-term memory assessments than those who didn't consume coffee (Raquel and Jesus). This suggests that volatile compounds used during the roasting of the coffee could enhance short-term memory which could potentially impact academic success of students by allowing them to recall information better. This is particularly relevant for high school students, who are often required to engage in tasks that demand high focus, such as studying for exams and completing assignments. It shows that although caffeine could allow individuals to feel more alert and improve memory in the short term, it doesn't fully compensate, compared to quality sleep.

While caffeine has been shown to enhance cognitive performance, particularly in short-term memory, there are some implications of caffeine on overall productivity. Raquel and Jesus (2016) found that short-term memory improved in groups exposed to coffee volatile compounds during testing, but did not show similar benefits for long-term memory after sleep deprivation. Showing that the impact of coffee could be limited for tasks that require sustained cognitive ability, including those that depend on long-term memory.

Consumers' Well-being and Health

In a study conducted by Ahsan F and Bashir S (2019), there is a significant correlation between caffeine consumption and decreased risk of Parkinson's disease and type 2 diabetes. This was further supported by Fischer et al. (2019), who found that coffee consumption has been associated with reduced risk of diseases like Alzheimer's and prostate cancer. Fischer et al. also emphasize that moderate consumption tends to bring health benefits, especially if consumed as plain black coffee, while excessive amounts could lead to adverse effects like risks for cardiovascular diseases and gastrointestinal problems.

Coffee consumption has been shown to also have psychological benefits, including reduced risk of depression and suicide (Quereshi et al., 2017). Through a meta-analysis with around 327,600 participants, they found that higher coffee consumption was linked to a 24% lower likelihood of depression (Quereshi et al). This is significant for high school students, who often face emotional challenges related to academic pressure and other personal stresses. The ability to manage these emotional challenges could potentially improve students' overall academic success. Furthermore, Pareek et al. (2024) found that coffee's stimulating effects could improve mood and reduce fatigue. As a result of having to balance studying, extracurricular activities, and other commitments, the positive effects of coffee on an individual's mood could provide relief and potentially improve productivity.

Gaps

Existing research on caffeine's effects on cognition and health has used methodological approaches, including observational studies, clinical trials, and meta-analysis. Many of these studies primarily focused on an older age range, leaving a significant gap in understanding how

caffeine affects younger individuals. Given the increasing prevalence of coffee consumption among high school students, further investigation is needed to determine how caffeine affects their learning and overall academic performance. Furthermore, many studies have focused on the general effects of caffeine on cognitive function, but few have examined how these effects translate to academic success. The researcher intends to fill this gap by focusing on the high school population and add a more in-depth understanding of the correlation of coffee intake from an educational environment and how it affects the academic success of NYC high school students. Filling this gap will advance the understanding of the cognitive effects coffee has on NYC high school students, and how certain factors play a role in it.

Hypothesis

Given this, it is hypothesized that moderate coffee consumption will positively influence the academic performance of high school students. This is shown in previous research suggesting that coffee enhances cognitive function, particularly in memory, reaction time, and attention. The researcher would conduct a quantitative study involving surveys distributed among high school students in NYC. The survey will measure students' coffee consumption and correlate it with academic success, as indicated by their academic performance and productivity levels. This study could inform dietary choices and clear up stereotypes on the relationship between coffee consumption and academic success. It will help educators and parents develop strategies to better support student learning.

Methodology

Purpose Statement

The purpose of this study is to investigate the relationship between coffee consumption and academic performance among high school students in New York City. Specifically, it explores how moderate coffee consumption influences students' GPA and productivity levels. By collecting data from a diverse sample of high school students, this study aims to provide insights into the cognitive benefits and potential risks associated with caffeine consumption among high school students. It will provide valuable information for educators, parents, and policymakers to develop better strategies to support students' academic success.

Research Design

This study used a quasi-experimental approach with a cross-sectional survey design to examine the cause-and-effect relationships between coffee consumption and academic success among high school students. A cross-sectional design allows for collection of data at a single point of time, making it ideal for measuring patterns and associations between coffee consumption and GPA. While a quasi-experimental approach allows for the comparison of two groups' responses: coffee consumers and non-coffee consumers. This design made it effective in exploring associations of variables in large populations (Quereshi et al, 2022). The numerical data was analyzed statistically on the relationship between caffeine intake and academic outcomes to be collected. This approach allows for precise measurements and allows for results to be generalizable. It also allows for the examination of relationships without manipulating the outcome and allows for exploring associations in large populations. It ensured ethical data collection while minimizing the risk of bias from external influences.

Population & Sample

The survey was distributed among students from a 9-12 NYC school, occurring for around two weeks. The school's population includes 26% Black or African American, 13% Hispanic or Latino, 35% Asian or Native Hawaiian/Other Pacific Islander, 22% White, and 3% Multiracial students. Gender distribution is 44% male and 56% female as of the 2022-2023 school year. Participants were informed of the study's purpose and the conditions for participating, with their consent obtained prior to the survey. Simple random sampling was used to select participants where classes were chosen to represent the broader school population. The survey was distributed based on a shared class and informed consent was obtained from students and teachers. This approach allowed for more efficient data collection and ensured that a diverse sample of students participated. Anonymous responses were used to protect their data and decrease bias in the results.

Instrumentation

The survey was created using Google Forms and depending on their level of coffee consumption, they would be given different sets of questions. The students were categorized based on their self-reported levels of coffee consumption: low coffee consumers (one cup or less per day) and high coffee consumers (more than one cup per day). This categorization allowed for the examination of the varying levels of caffeine intake and impact on function (Santos et al., 2024). Students who did not drink coffee were given a different set of questions to compare relevant data without introducing variables that could skew the results.

The results were selected as a sample population to represent a diverse group. The demographic of the sample was not specifically controlled for factors like gender or ethnicity, as

the focus was on coffee consumption and academic performance, regardless of these variables. The survey included both quantitative questions (such as GPA and coffee consumption frequency) and qualitative open-ended questions aimed at gathering additional context about study habits, perceived focus, sleep, and other variables that might contribute to a difference in data. Informed consent was obtained from all participants before they completed the survey. Participation was entirely voluntary, and students were informed that their responses would remain anonymous and confidential. Participants also had the option to skip questions they weren't comfortable answering.

Procedure

After the data were collected, the data was visualized using Python, seaborn, pandas matplotlib on Google Collab. Box plots were created to compare the GPAs of low vs. high coffee consumers, and a t-test to find p value was conducted on the qualitative data to uncover recurring patterns related to study habits and focus. The data was then analyzed to significant patterns between the relationship between coffee consumption and academic performance, as well as productivity levels.

Data Analysis

The data collected from the survey were transferred into Google Colab, allowing for more in depth data analysis. This platform allowed for more data manipulation and visualization through Python. The researcher used tools such as Pandas for data manipulation and analysis, to prep the data. Any missing or incomplete data was cleaned up, ensuring all responses were categorized correctly for further analysis. It was also used to filter the data based on consumer and non-consumer groups. Two Python libraries, Matplotlib and Seaborn, were then used to

create visual representations of the data. The researcher mainly created boxplots to examine the distribution of the variables and identify if there was any significant data. Box plots were ideal to display the distribution of responses, including the interquartile ranges, median, and potential outliers in the dataset. It also allowed the researcher to compare the consumer and non-consumers responses side-by-side for easier comparison. The visualization helped to reveal significant trends and variations in patterns between the two groups. Additionally, to assess whether the differences between the two groups were statistically significant, t-tests were performed. The p-value was calculated through Python's statistical library SciPy which was also displayed at the bottom of each graph. The p-value represented the strength against the null hypothesis, with a lower p-value (below 0.05) suggesting that the difference is statistically significant and that the data were unlikely to be due to random chance.

After analyzing the quantitative data gathered from the survey, the researcher combined the quantitative data from the survey to find a connection to create a more in-depth conclusion on the relationships between coffee intake and academic success of NYC high school students. The researcher also examined the qualitative responses from the open-ended survey questions which provided additional context to the trends in the responses which was subsequently considered in the analysis.

Results

Surveys

The survey collected data from a 9-12th high school in NYC, to examine patterns in coffee consumption and how it may influence academic performance and productivity. To analyze the data, unpaired t-tests were conducted comparing coffee consumers with non-coffee

consumers across several metrics: GPA, perceived productivity, perceived academic performances, and sleep patterns. Box plots were used to visually present differences between the two groups, allowing for a more clearer understanding of the distribution of responses.

Figure 1: GPA Comparison Between Consumers and Non-Consumers

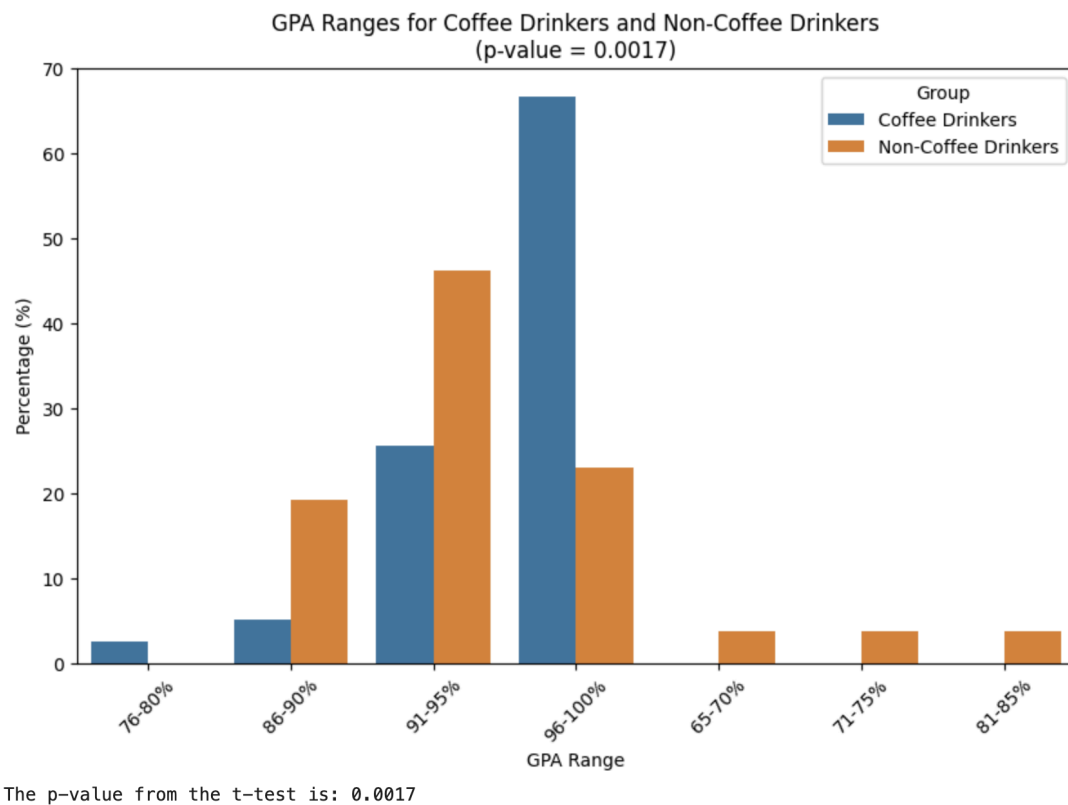


Figure 1 shows that non-coffee consumers generally had a higher GPA than coffee consumers. The distribution of GPA scores among non-consumers was skewed toward the upper end of the spectrum. The p-value was statistically significant ($p < 0.05$), suggesting a significant association between the two groups in terms of GPA.

Figure 2: Sleep Patterns Between Consumers vs. Non-Consumers

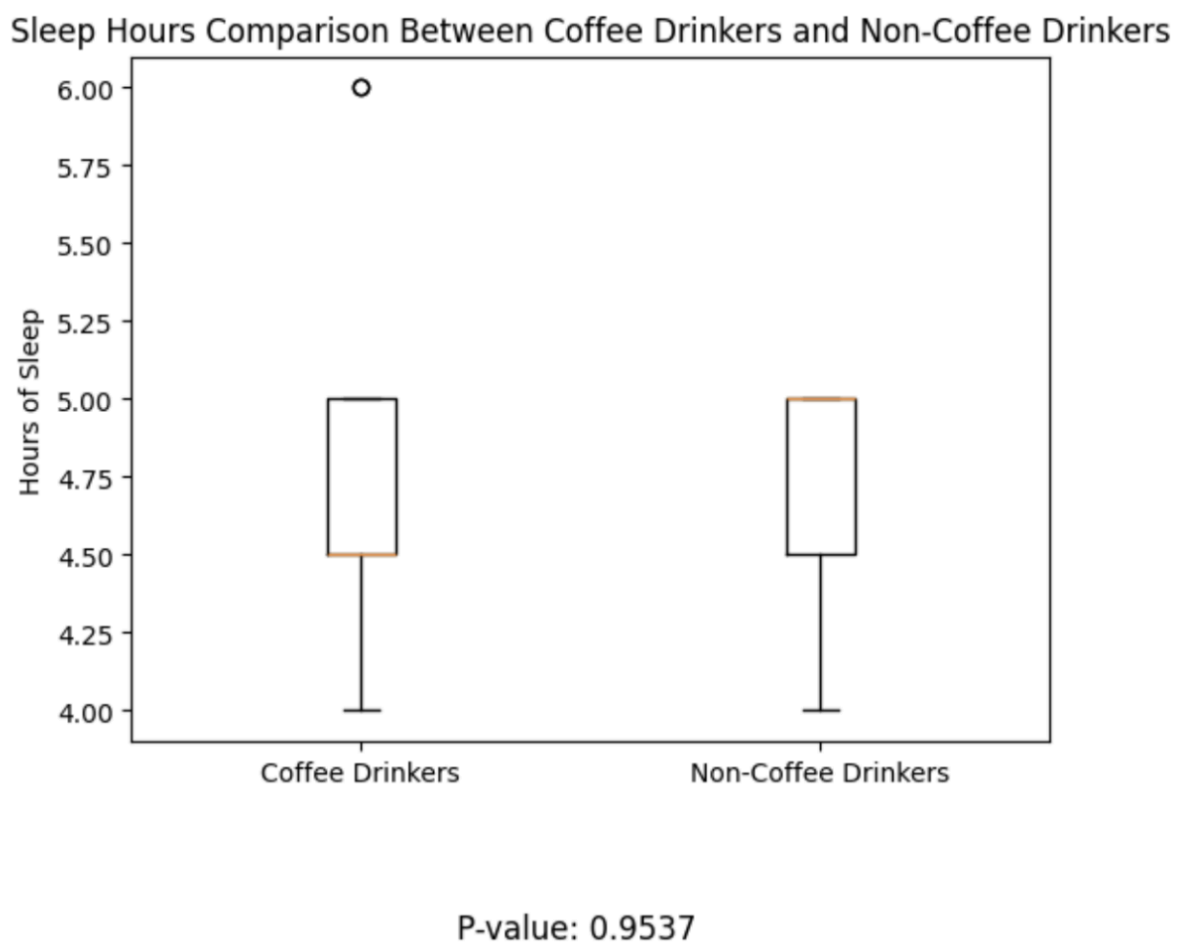


Figure 2 shows no significant difference in sleep duration was observed between coffee consumers and non-consumers ($p < 0.05$). Both consumers and non-consumers reported similar sleep ranges, with the p-value being 0.9537, well above the 0.05 threshold. Suggesting that, coffee consumption is not necessarily associated with reduced sleep.

Figure 3: Perceived Academic Performance After Consuming Coffee

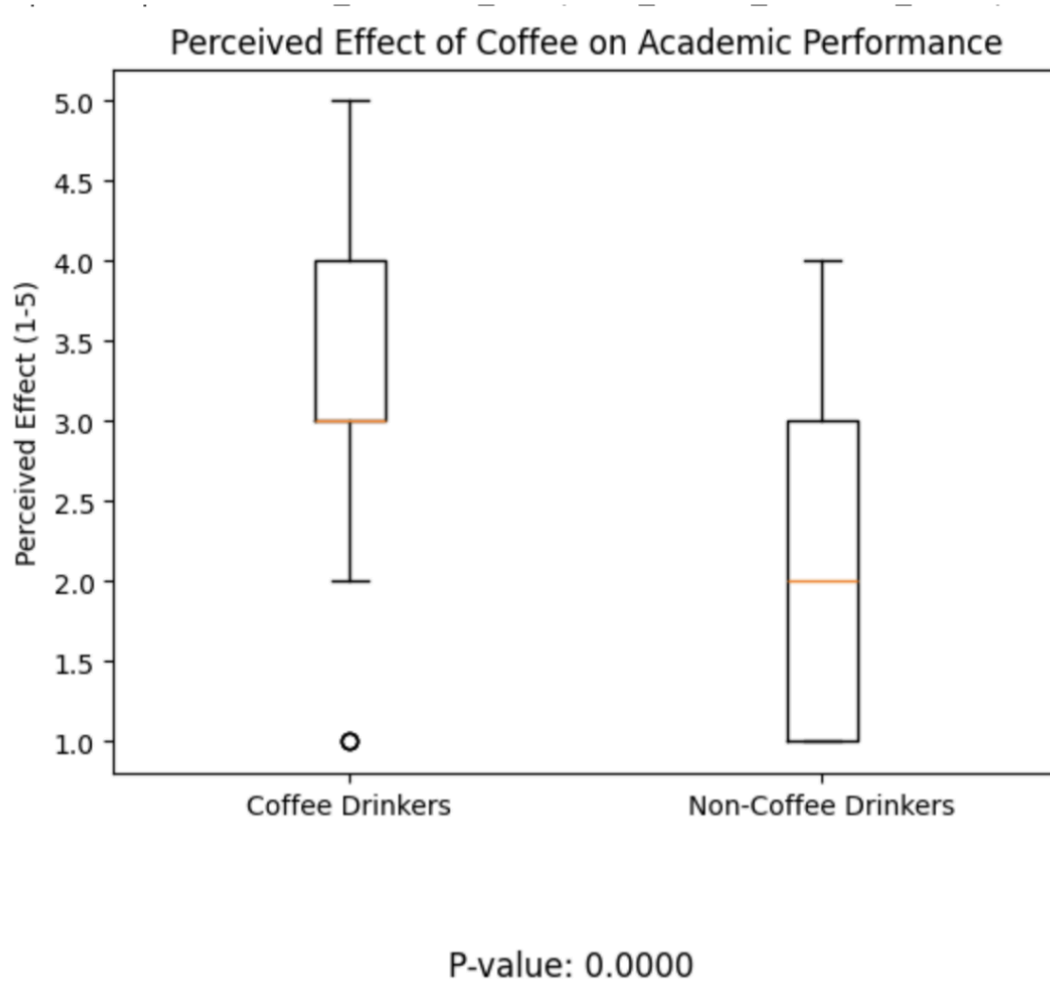


Figure 3 shows that students' perception on how coffee affects their academic performance. A significant number of consumers reported that consuming coffee helps them perform better academically. The t-test revealed a p-value of 0.0 which suggests a strong statistical significance.

Figure 4: Perceived Productivity Levels After Consuming Coffee

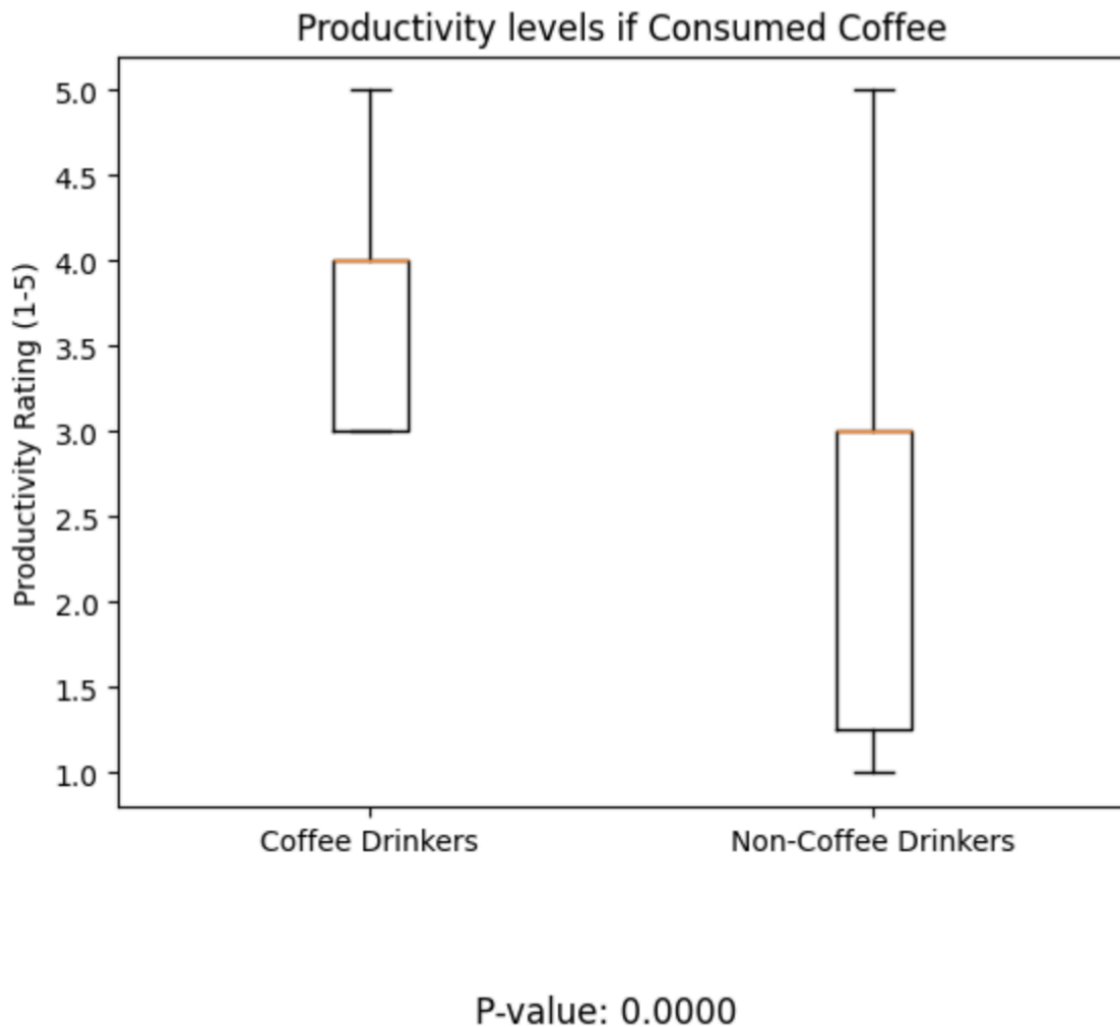


Figure 4 shows students' perception on how coffee influences their productivity levels. It shows that consumers reported higher productivity levels compared to non-consumers. The t-test for productivity yielded a p-value of 0.0000, suggesting a significant correlation between coffee consumption and perceived productivity.

Findings & Discussion

The findings show evidence for a possible relationship between academic performance and productivity levels in high school students. Specifically, coffee consumers reported higher productivity levels compared to non-coffee consumers. This suggests that moderate coffee consumption may play a role in boosting cognitive function, especially in academic settings. The connection can be shown in Figure 1, where coffee consumers predominantly fall in the 91-100% GPA range, whereas non-coffee consumers have a wider distribution, with most falling into the 96-100% GPA range. This discrepancy suggests that there might be a correlation between consuming coffee and GPA, with a p-value of 0.0017 which is lower than the threshold 0.05.

Figure 2 supports this by showing that consumers report higher productivity ratings compared to non-coffee consumers. With a t-test of 0.0000, it shows that coffee consumption could play a role in boosting productivity and enhancing cognitive function. This supports the hypothesis that caffeine can enhance productivity by promoting mental awareness. This aligns with Santos et al. (2010) who concluded that moderate caffeine intake can improve memory, focus, and reaction time, which are crucial cognitive abilities for academic tasks. This is significant as it shows the impact of coffee on academic performance among students.

However, there is no significant difference in sleep patterns between consumers and non-consumers, as seen in Figure 2. Both groups show similar average hours of sleep, supported by the p-value of 0.9537 from the t-test. Since coffee consumption is often linked to sleep disturbances in some individuals, this result challenges that finding. Haskell-Ramsay et al. (2018) found that excessive caffeine consumption can interfere with sleep, which in turn impacts

cognitive performance. However, it shows that sleep habits do not affect coffee consumption and do not play a role in the effects on academic performance and productivity levels.

The researcher's hypothesis was partially correct, as shown in Figures 3 and 4, where consumers showed differences in academic performance and productivity levels when consuming coffee versus not consuming coffee. However, the hypothesis that coffee consumption could affect sleep patterns was not supported, as Figure 2 showed that there is no significant difference between the two groups. The lack of a clear relationship between coffee consumption and sleep duration requires other factors to be taken into account, such as whether the respondents consumed other caffeinated beverages. This was also measured in the survey, as these drinks can also have the same effects as coffee, which may impact students' academic performance and productivity levels. In addition, when asked why non-consumers do not drink coffee, 51.7% of respondents indicated that they prefer other beverages, 23.3% cited not liking the taste, and others provided various reasons. The findings revealed that a significant portion of non-coffee consumers also regularly consume other caffeinated beverages. For example, some participants reported drinking energy drinks, tea, and soda, either as alternatives to or in combination with coffee. This preference could be driven by several factors, including taste, habits, or a general lack of interest in coffee. Overall, the analysis supports the hypothesis that coffee consumption is associated with higher productivity and academic performance.

New Understandings

After analyzing the data collected from the survey, this study fills the gap in understanding the relationship between coffee consumption and the academic performance and productivity of high school students in New York City. Initially, the researcher hypothesized that

moderate coffee consumption would have a positive effect on students' academic outcomes, especially their productivity and GPA. The findings confirm that there is indeed a positive correlation between coffee consumption and higher productivity levels, with coffee consumers reporting a more significant sense of alertness and perceived academic success. However, the study also revealed that while coffee consumption was linked to better perceived productivity and academic performance, it did not affect sleep patterns. The lack of a significant relationship between coffee consumption and sleep duration challenges the assumption that coffee negatively impacts sleep and thus academic outcomes. Therefore, this study contributes to a nuanced understanding of how caffeine consumption impacts students' academic performance, emphasizing the need for further investigation into other factors, such as the consumption of alternative caffeinated beverages, that might influence these outcomes.

Limitations

This study has a few limitations that should be addressed in future research to ensure better understanding of the relationship between coffee consumption and academic performance. For instance, the sample population was primarily drawn from a single high school, which may not adequately represent the diversity of students across all New York City schools. The results may differ if the sample included students from a wider range of schools with varied socio-economic backgrounds which could influence coffee consumption and academic performance. Furthermore, this could limit the ability to generalize the findings. Additionally, the reliance on self-reported data could cause biases with over-reporting or under-reporting coffee consumption or academic performance. Students may over-report or under-report their coffee consumption, either due to bias or not being able to accurately track or recall their consumption.

These biases could potentially skew the data, leading to inaccurate conclusions about the true relationship between coffee consumption and academic performance.

Implications

This research provides valuable insights into the relationship between the academic success of NYC high schools students. By looking into how coffee consumption affects students' academic performance and productivity, the findings can contribute to fields of education, health, and child development. Additionally, could also help educators, parents, and researchers to help students' make better choices that align with their academic goals and overall well-being.

Future Directions

Future research should focus on expanding the sample size to include students from different grade levels and schools across a broader range of demographics. By including a more diverse group of participants, researchers could examine the relationship between coffee consumption and academic performance with greater accuracy. Additionally, exploring the impact of other caffeinated beverages, such as energy drinks, tea, or soda, on academic performance and productivity could offer a meaningful comparison to better understand how coffee compares to other caffeinated drinks on effects on students' academic outcomes. An experimental approach that tracks students' performance on assessments after consuming caffeine is recommended. This could help determine whether time of consumption and sleep patterns influence academic results. However, due to limited resources and inability to conduct such an in-depth experiment, the researcher was unable to investigate these variables further.

Conclusion

This research explored the relationship between coffee consumption and its perceived effects on productivity and academic performance among high school students in NYC. The findings suggest that coffee consumption is associated with increased levels of perceived productivity and academic performance. These results support the hypothesis that caffeine intake, in moderation, may play a role in improving cognitive function and academic motivation in high school students.

However, the results did not show a significant correlation with sleep patterns. Both group showed similar sleep duration and quality, with and without the presence with coffee. This challenges common assumptions about caffeine's impact on sleep quality. Other variables may have also influenced the results. Several participants reported consuming other caffeinated beverages like energy drinks, soda, or tea in place or in addition to coffee. These alternatives could contribute to similar cognitive or productivity-related outcomes, making it difficult to isolate the effects from coffee. Similar to previous research on the effects of decaf coffee being similar to that of regular coffee, its consumption may have influenced the results.

Ultimately, this study contributes to the growing understanding of how factors like coffee consumption may shape the academic success of high school students. While caffeine could be a useful tool for improving focus and cognitive ability, further research on its long-term effects is necessary to draw more definitive conclusions.

References

- Cano-Marquina, A., Tarín, J. J., & Cano, A. (2013). The impact of coffee on health. *Maturitas*, 75(1), 7–21. <https://doi.org/10.1016/j.maturitas.2013.02.002>
- F, A., & S, B. (2019, July 1). *Coffee consumption: Health perspectives and drawbacks*. <https://www.scholarena.com/article/Coffee-Consumption-Health-Perspectives-and-Drawbacks.pdf>
- Farias-Pereira, R., Park, C. S., & Park, Y. (2019). Mechanisms of action of coffee bioactive components on lipid metabolism. *Food Science and Biotechnology*, 28(5), 1287–1296. <https://doi.org/10.1007/s10068-019-00662-0>
- Fischer, E. F., Victor, B., Robinson, D., Farah, A., & Martin, P. R. (2019). Coffee: Consumption and health implications. In A. Farah & A. Farah (Eds.), *Coffee: Consumption and health implications* (pp. 1–19). The Royal Society of Chemistry.
- Haskell-Ramsay, C. F., Jackson, P. A., Forster, J. S., Dodd, F. L., Bowerbank, S. L., & Kennedy, D. O. (2018). The acute effects of caffeinated black coffee on cognition and mood in healthy young and older adults. *Nutrients*, 10(10), 1386. <https://doi.org/10.3390/nu10101386>
- Loftfield, E., Freedman, N. D., Dodd, K. W., Vogtmann, E., Xiao, Q., Sinha, R., & Graubard, B. I. (2016). Coffee drinking is widespread in the United States, but usual intake varies by key demographic and lifestyle factors. *The Journal of Nutrition*, 146(9), 1762–1768. <https://doi.org/10.3945/jn.116.233940>

- Pareek, V., Shah, S., Sharma, B., Kumar, S., & Sharma, L. (2024). Coffee and the brain: A comprehensive review of its neurological and psychiatric effects. *Journal of the Indian Academy of Geriatrics*, 20(1), 34–39. https://doi.org/10.4103/jiag.jiag_53_23
- Qureshi, F., Stampfer, M., Kubzansky, L. D., & Trudel-Fitzgerald, C. (2022). Prospective associations between coffee consumption and psychological well-being. *PLOS ONE*, 17(6), e0267500. <https://doi.org/10.1371/journal.pone.0267500>
- Raquel, Q.-O., & Jesús, M.-E. (2016). Effect of the volatile coffee components on memory in sleep deprived students. *Revista Mexicana de Neurociencia*, 17(2), 42–48.
- Santos, C., Costa, J., Santos, J., Vaz-Carneiro, A., & Lunet, N. (2010). Caffeine intake and dementia: Systematic review and meta-analysis. *Journal of Alzheimer's Disease*, 20, 187–204. <https://doi.org/10.3233/JAD-2010-091387>
- Viado, M. P. (2024). Coffee consumption and its perceived effects on the study habits of higher education students. *International Journal of Multidisciplinary: Applied Business & Education Research*, 5(8), 3062–3070. <https://doi.org/10.11594/ijmaber.05.08.11>